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Design and Development of Integrated Insole System for Gait Analysis

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***Abstract*— An insole system has been developed in this research, which is helpful in acquisition and analysis of gait parameters. Daily auditing is necessary to evaluate the quality of gait before, during and after rehabilitation. Footgear systems have been quite a popular choice for researchers as they are non-invasive with user friendly setup. The designed device analyses gait patterns using FSR type pressure sensors integrated within the shoe insole. The location of the sensors were identified by taking the footprint of the subject in a sandbox and then marking the depressions. The data was logged in each step and parameters were extracted through post processing of the acquired data. Experiments were performed to validate the reliability of the acquired data. The data showed symmetry for a healthy subject in terms of similar step length, stride length and velocity. A wireless pressure sensor-based insole system has been developed to monitor gait parameters and critical events. The device was used to calculate different parameters of one subject for 4 trials with different terrain/path lengths of 10 meter, 12 meter, 14 meter and 16 meter. The parameters worked out in this study were average stride length as 1.20±0.09 m, average step length as 0.60±0.05 m and average velocity as 0.67±0.03 m/sec along with heel strike and toe-off events for all trials.**

Keywords*— *Wireless Communication, Insole, Pressure Sensors, Gait Characterization, Data Processing.

I. INTRODUCTION

GAIT anomalies are usually associated with varied medical cases such as tumors, congenital causes, trauma, Parkinson's disease, Cerebral Palsy, Huntington's Disease etc. [1]–[3]. Other research areas are being explored to prevent Plantar Fasciitis [4] (ankle pain) due to micro tears and Metatarsalgia [5] (front foot pain) for reasons of overpronation among professional runners and percentage of such cases is on the rise during the last decade. Injury prevention and performance enhancement are a few amongst many objectives targeted in this research. Other purposes include the diagnosis, analysis and treatment of neurological and musculoskeletal disorders [6] associated with the interaction of normal forces during swing and stance of support surface and foot. In certain research articles locomotion identification using sEMG signals has been utilized to design lower limb prosthetics [7]-[9]. This gait analysis will be helpful in designing new and advanced versions of existing footwear models.

The existing wearable technology provides very limited aid in understating the mechanics of running thereby contributing very little in injury prevention. Thus, in order to fully understand the mechanics the experiments are being performed in motion capture laboratories. To investigate the pattern of walking/running, gait analysis was earlier done in two ways viz., the motion lab measuring the parameters with high resolution optical tracking and the second one being the visual observation.

The first method i.e., motion lab measuring the parameters, although, produces highly accurate and well quantified results, but is uncomfortable and cumbersome and very expensive too. The second method, visual observation is highly subjective in nature thereby leading to poor reliability, sensitivity, doubtful validity in comparison with gait analysis with instruments [7], [8]. In the optoelectronic systems [9][10] the data sampling rate vary from 50-1000 Hz and that a 60 Hz sampling rate is sufficient for the gait analysis of a subject rather [11] suggests that most of the body movements are well restricted to a frequency range of 15 Hz or low. Another aspect of gait monitoring is the footwear as it behaves as an interface between foot and ground. The applications areas of footgear-based strategies vary from step counting to advanced systems utilized during rehabilitation courses for physically disabled subjects. The systems available in research laboratories or market vary in the sensor techniques and data acquisition methods to cater to varied application necessities.

The device developed here is used to monitor gait parameters such as stride velocity, step length etc. using pressure sensors at different locations and then transmitting them to data acquisition device using RF communication protocol. The device was worn on both the feet and another device was connected to computer to receive data. Post processing of the data is done on computer and parameters are extracted using MATLAB as the data processing software.

II. DEVICE DESIGN/ SYSTEM CONFIGURATION

A. Material & Methods

The data acquisition systems mentioned above, consist of pressure measurement sensors located on the plantar side of the foot, inertia measurement sensors such as accelerometer, gyroscope for motion detection and different protocols for communication and data acquisition. Further processing of the acquired data can be done according to the application. Various crucial biomechanical parameters can be computed via placing sensing elements in the insole of footgear. Moreover, with

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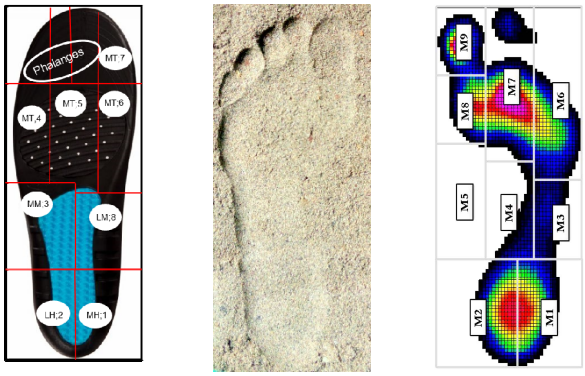
inertia measurement sensors, more gait parameters such as acceleration, orientation can be measured. In addition to this, an actuator can be placed inside the footgear, in which biofeedback can be generated, thereby, aiding subjects suffering from stroke. Further, activity recognition, energy expenditure and posture can be tracked. These are few of the various applications, which act as the driving force behind the innovation in the footwear technology. The system has to ensure that data collection is always done correctly so that the quality of dataset is reliable & good and the results based on this dataset are accurate. Other aspects such as data analysis and feature extraction are being done offline.

The system designed here is scalable i.e., it can be used indoor and outdoor without any complication. The device is standalone and can be used with other electronic devices in surroundings without any interference. In-terms of usability and portability, the device is non-invasive, easy to wear and require minimum training. The timing and pressure parameters were accessed via the Force Sensitive Resistors (FSR) sensors placed under the foot on the insole. FSR of diameter 12.7 mm were used, manufactured by Interlink Electronics [12]. Out of nine [13] anatomical sub-areas, eight were considered to measure pressure only from relevant locations. The location of sensors can be seen in Figure 1(a). The locations of the sensors was variable in the areas indicated by the red lines.



(a) (b) (c)

Figure 1: Sensor Localization (a) Sensor locations in current device (b) Sandbox Footprint (c) Anatomical subareas

Sensor movement was allowed so that the insole can fit to people with different feet sizes. Each subject was asked to stand barefoot in a box of sand and the areas, where the sand was most compressed, were the areas chosen for placing the sensors. This way, a high amount of customization can be provided for subjects with varied foot physiology including size, shape etc. The insole was fully covered with Velcro, to allow the movement of the sensors depending on the subject's feet size and pressure points. The data being acquired was received by Arduino Nano which has an Atmega328 microcontroller embedded [14]. The analog pin received the input and converted it into a corresponding digital value ranging from 0 to 1023 by a 10-bit inbuilt ADC. Before reaching the input to MCU, the eight data lines from eight pressure sensors are routed via 8x1 multiplexer [15].

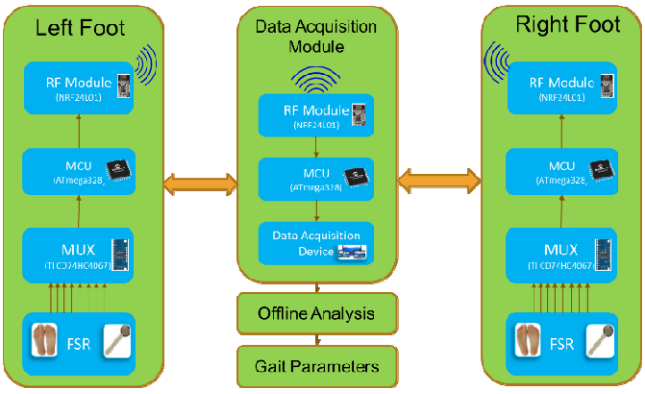
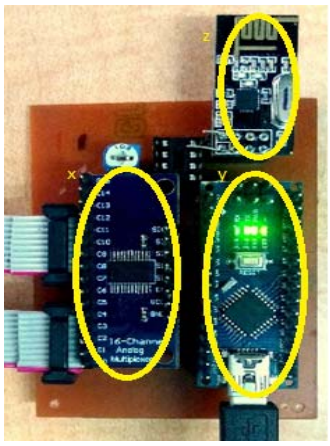


Figure 2: Device Block Diagram, The Left Foot and Right Foot Data Transmission Modules are identical and the Data Receiving module is connected to computer where the offline Analysis of data is done to extract gait parameters

Insole was inserted inside each sandals/shoes and the data acquisition device was strapped on lower legs of the subject. Both the data acquisition devices have a transmitting module, to transmit data acquired, to a master module connected to the computer. The communication between master module and slave modules was bidirectional communication and polling method was followed. The master module asked for data to each of the slaves by sending a predefined data bit. If the bit was successfully received, data was sent to master. The master then asked the second slave to send data and this procedure was repeated for the whole experiment duration. The received data was being logged into the computer with time stamp corresponding to each reading. The module responsible for wireless communication was nRF24l01 RF module from Nordic Semiconductor [16] which works in 2.4 GHz ISM band. Both the slaves and the master module were communicating through this module. The slaves had individual addresses and responded only when they were being referred to. The complete data transmission module is shown in Figure 3(a).



(a) Data Transmission Module

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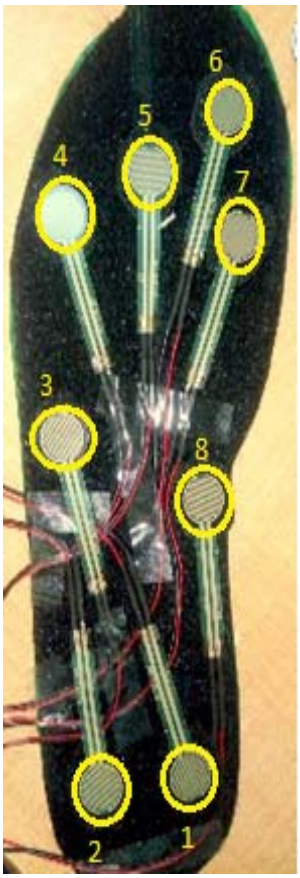

(b) Internal Construction of Insole

Figure 3: Device Construction (a) (x) Multiplexer Module (y) Microcontroller Module (z) Nrf24l01 RF Module (b) Insoles Embedded with Force Sensors (1) Medial Heel (2) Lateral Heel (3) Medial Midfoot (4) Metatarsal Head 1 (5) Metatarsal Head 2 (6) Hallux (7) Metatarsal Head 4 (8) Lateral Midfoot

In Figure 3(b) the location of sensors can be clearly seen. Sensor 1 and 2 correspond to detection of heel strike, Sensor 3 and 8 correspond to detection of direction of motion of foot. Sensor 8 can detect physiological disability of Flat Foot and Sensor 4, 5, 6 and 7 detect parameters corresponding to forefoot which include detection of disabilities such as Lateral Whip and Medial Whip [17] and toe off timings.

B. Parameters measured

Spatio-temporal parameters like stride length, step frequency, cycle, time, stance time etc. were calculated using the embedded pressure sensitive elements. The various parameters were easily revealed using the heel strike and toe off time instances. The information provided by the sensors is differential change in pressure instead of absolute change (unless calibrated), therefore, ultra-high resolution is not needed for most of the parameter estimation.

Step length and stride length are some of the important parameters for gait analysis however the illustrated device is capable of calculating other important parameters like Velocity, Centre of Pressure, Double support Percentage, Single Support Percentage, No. of Steps.

a. Stride Length (m)

The stride length is the distance between the previous heel strike of the right foot and the new heel strike of the right foot. It can be clearly seen in Figure 4.

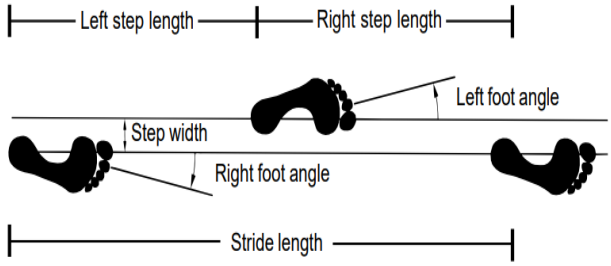

Figure 4: Stride Length and Step Length Description

b. Step Length (m)

The step length is divided into step length of right foot and step length of left foot. The step length of right foot is distance between previous heel strike of left foot and the next heel strike of right foot and similarly step length of the left foot. Ideally both the step lengths should be almost equal and if there is deviation greater than a certain degree than it might be due to some abnormality which needs to be addressed as soon as possible.

c. Velocity (m/sec)

This parameter was calculated by fixing the path length for the subject under experimental conditions. The average velocity was calculated by dividing known distance by total time taken.

d. Single and Double Support Percentage

Single support percentage refers to the amount of time the subject was on one foot only. It can be measured starting from toe off of one foot till the heel strike of the same foot. Similarly, double support percentage was the amount of time the subject is on both the feet. It can be calculated by subtracting the single support time from stride time.

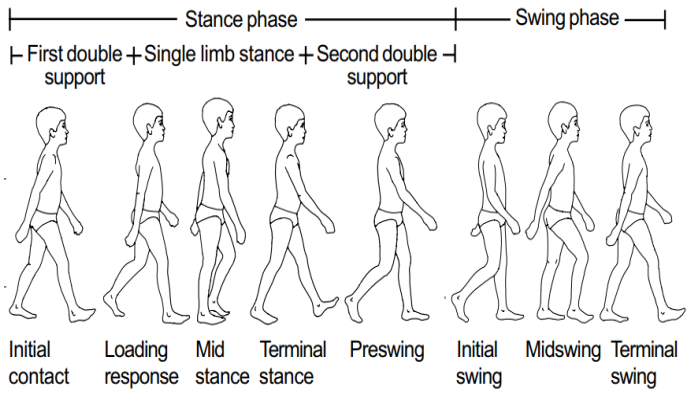

Figure 5: Complete Gait Cycle

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III. EXPERIMENT PROTOCOL TO BE FOLLOWED

A protocol was set before any kind of experimentation using the device was done on the subjects. A long corridor was chosen as the experimentation area. The following steps were followed in order for the experiment to be performed.

- The subject was asked to stand in the sandbox barefoot.
- Necessary adjustments were made to the sensor locations.
- The subject was then made to wear sandals with insole system with or without socks.
- The subject was asked to get comfortable with the new device, by taking a walk at a normal pace till he feels comfortable physically as well as psychologically, which if not done, may lead to degraded dataset.
- Finally, the experiment started and the subject was asked to walk normally in walkway/ corridor to and from a specifically marked point.
- Thereafter, the subject was asked to carefully remove the sandals for the next subject.

The subject chosen for the initial prototype testing was a healthy subject, without any kind of physiological and psychological disability. During the walk, data was being recorded in the computer and when the experiment was complete the dataset was saved using the name of the subject. Apart from the data, certain demographic parameters were also acquired by asking the subject to fill out a form, which includes details like age, height, weight etc. This data will be useful to find a certain correlation between the various gait parameters and demographic data.

Figure 6 shows the complete device for a single foot and Figure 7 shows the subject wearing the instrumented insole with data acquisition module. All the experiments were performed with this setup only and the results have been described in Table 1.

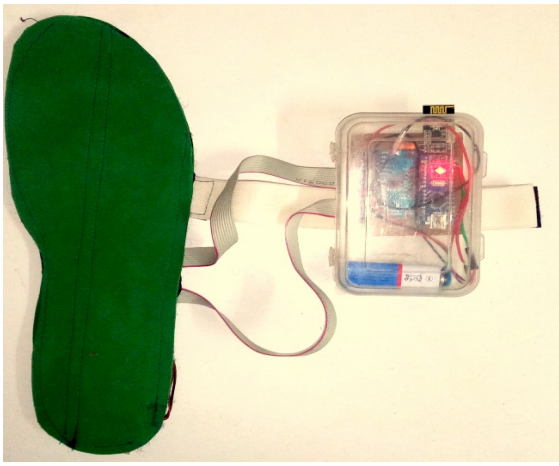


Figure 6: Insole with Data Transmission Module (Complete Setup). The Sensors described in Figure 3(b) are covered with another thin flexible insole to provide the subject a feeling of comfort and to prevent any damage to sensors as well.

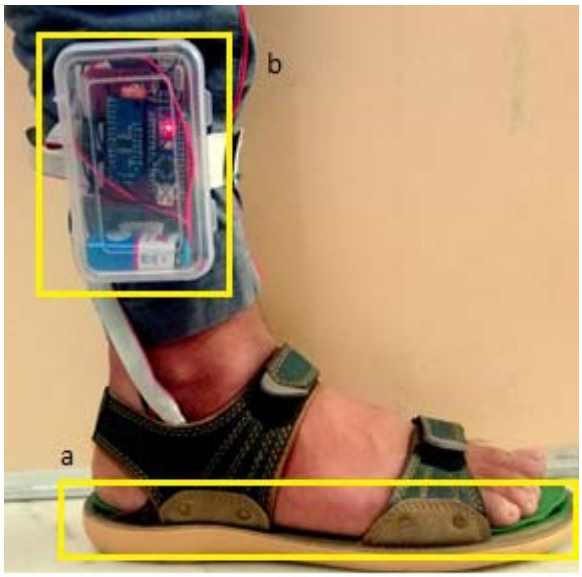
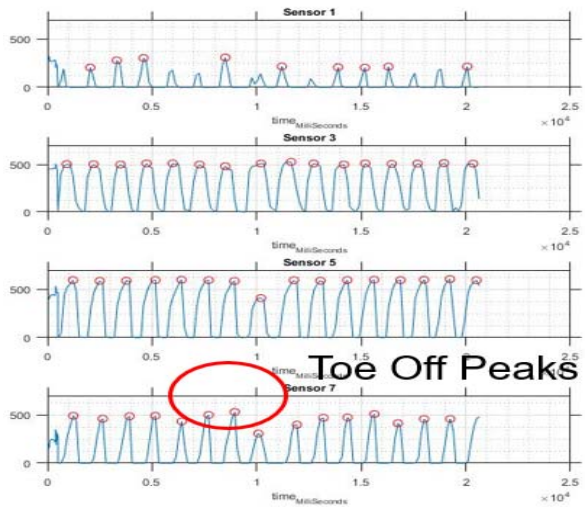


Figure 7: Subject with Wearable Module: (a) Instrumented Insole (b) Data Transmission Module

IV. RESULTS AND DISCUSSION

To validate and evaluate the system performance, certain parameters were analyzed which included average stride length, average step length, average velocity and step count. Walking pattern/gait of a pedestrian can be analyzed by observing the aforementioned parameters.

The graphs described in Figure 8 correspond to the eight sensors, graphs 1 and 2 for heel areas and so on. The y-axis represent the digital value corresponding to the force applied on the sensor and the x-axis represent the time in milliseconds. No peaks in the eighth sensor data clearly shows that the person is not suffering from flat foot.



(a) Data Plots for sensors 1, 3, 5 and 7

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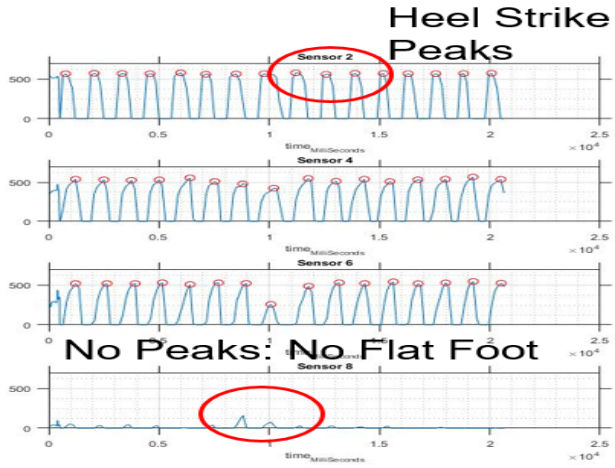

(b) Data Plots for sensors 2, 4, 6 and 8

Figure 8: Data Visualization for Each Sensor.

The prominence of peaks in the graph 2 corresponds to heel sensor 2, which implies that the subject's foot comes in contact with the ground with the right side of heel first. They can detect the cases of overpronation and oversupination.

TABLE 1. GAIT PARAMETERS

Subject	Distance Travelled (m)	Stride Length (m)	Step Length (m)	Avg. Velocity (m/sec)	Stride Count
Trial 1	10	1.0965	0.52	0.6202	10
Trial 2	12	1.3333	0.665	0.6821	9
Trial 3	14	1.2507	0.635	0.6895	11
Trial 4	16	1.1429	0.596	0.6966	14

Four trials were performed on the same subject, using the insole designed with different track lengths of 10, 12, 14 and 16 meters respectively. The average stride length calculated came out to be 1.20±09 m, average step length of 0.06±0.054 m and average velocity of 0.67±0.054 m/sec. The stride counts for the trials were also calculated and came out to be 10, 9, 11 and 14 strides for 10, 12, 14, 16 meters respectively. The heel strike and toe off events were also detected in the algorithm.

V. CONCLUSION

Gait characterization plays an important role in rehabilitation, lower limb prosthesis control, robotic control and pre-operative disease quantification. In this study, a low cost wireless pressures sensor based insole has been developed to capture change in foot pressures during locomotion. The location of sensors has been selected strategically as each location possess unique information regarding gait cycle. A preliminary study has been carried out, to justify the efficacy of developed device, by acquiring data of one subject for four trials at different terrain/path length.

The basic parameters like stride length, average velocity etc. has been calculated successfully from acquired signal along with critical gait events. The promising performance of developed device, justify its usability towards any application, where gait characterization plays an important role such as Parkinson's disease, cerebral palsy etc. Further the usability of the developed device can be increased, by incorporating orientation based sensors that will reveal more insights in the information of gait cycle.

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